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DEALER BEST PRACTICE SERIES

Improving Motor Grader Edge Installation Process with Special Fitting Tool

Application **Maintenance** Site Management Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

Motor grader edge replacement is typically a slow, labor-intensive, and ergonomically challenging process.

One dealer designed and fabricated a tool to hold the edge at the proper height and angle for improving installation operations. Using the tool has lowered the time required to install cutting edges (reduced downtime), lowered costs and labor, and minimized the manual handling required by the operator.

This Best Practice is a Quick Win project.

2.0 Best Practice Description

The edge installation tool shown in this Best Practice was designed and fabricated by the dealer. The key features are:

- The tool holds up to four edges plus the hardware needed to fit the edge to the machine (Figure 1).
- The edge is mechanically held at the correct angle. This allows the edge to be fitted without the use of manual handling (Figure 2).

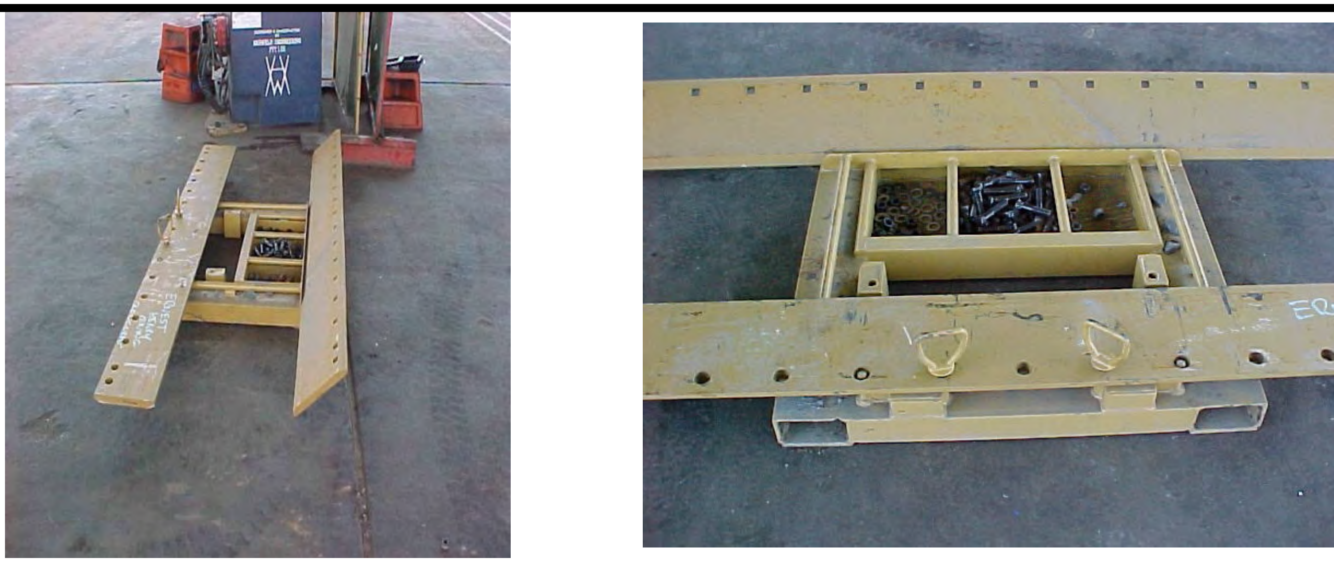

Figure 1 – MG fitting tool

A set of four edges is loaded onto the installation tool, which is then lifted by a forklift to the motor grader. Technicians install the edge using hardware being stored in the tool. After installation, the forklift simply moves down to the next position. Minimal handling of the edge is required.

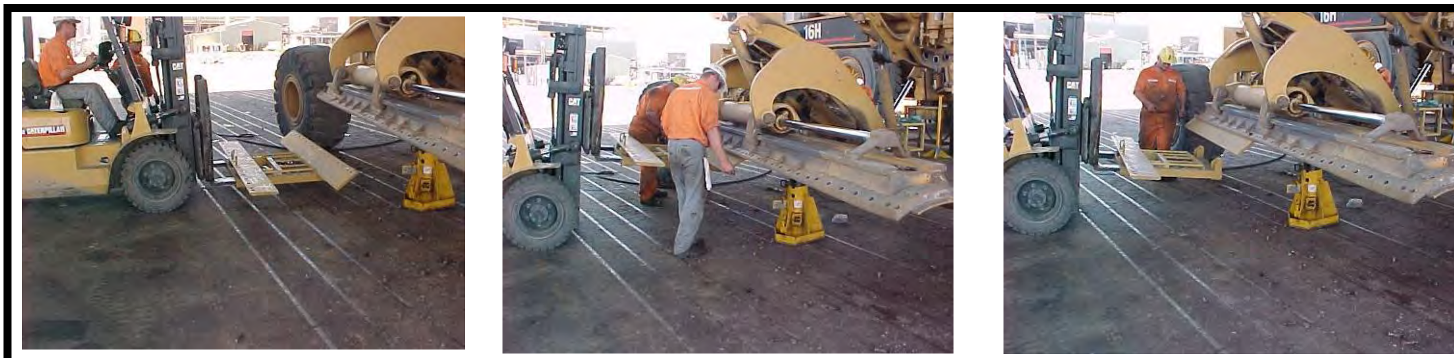

Figure 2 – Fitting tool in use

3.0 Implementation Steps

- Establish a need for the tool (risk assessment), including expected volume of motor grader edge replacements
- Purchase required steel for fabrication (consider size of motor graders)
    - o Fabrication/ welding crew can provide recommendations based on attached diagrams
    - o Use a sample set of edges from parts stock for fabrication measurements
- Collaborate with the Maintenance team (end users) to establish a fabrication design
- Fabricate the tool
- Based on a Job Safety Assessment (JSA), train users and implement the tool in the new process

4.0 Benefits

- Reduced installation time / cost, and reduced Mean Time to Repair
- The job only requires two technicians instead of three, saving considerable labor costs.
- Less manual handling by technicians, improving ergonomics and safety.
- If the motor grader is covered in a MARC, the labor costs saved during the edge installation process help to reduce dealer costs and contract risk.

5.0 Resources Required

- Fabrication material and time
- Attached diagrams
- For further instructions please contact Hastings Deering at Ernest Henry Mine, North West Queensland Australia (see Acknowledgements).

6.0 Supporting Attachments / References

Dimensions are shown in Figure 3. These dimensions are relevant for a 16H size motor grader. Enlarged photos are available in the attached PowerPoint file.

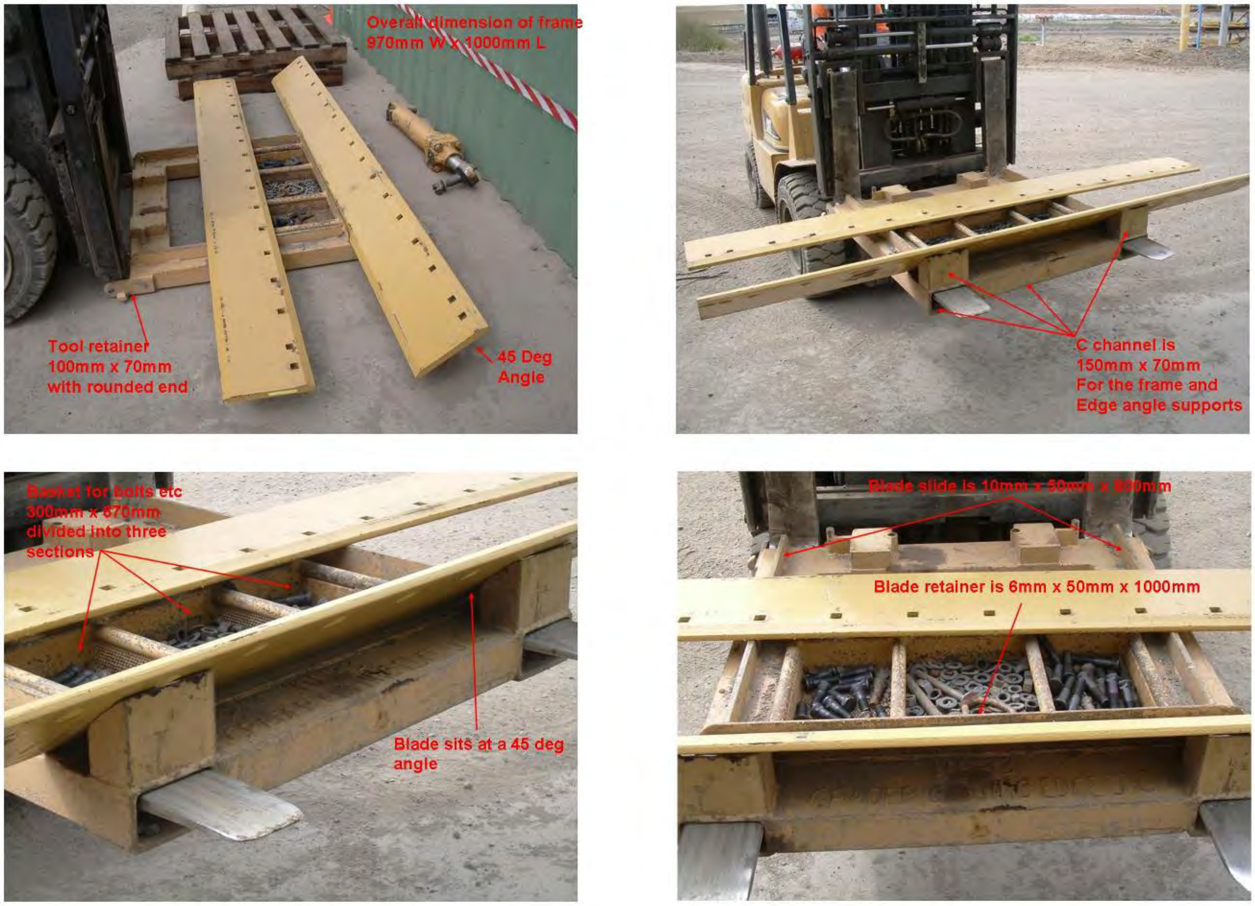

Figure 3 – Tool overall dimensions and specifications

7.0 Related Best Practices

While the Best Practice listed below relates to the Component Rebuild Center, and not specifically to Maintenance processes, the topic of labor efficiency and the related benefits serve as complimentary information to that presented here.

0207-4.5-1062 Labor Efficiency Lowers CRC Costs

8.0 Acknowledgements

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